

Performance Testing

Date	12 August 2026
Name	Chahat Chouhan
Team ID	Individual Project
Project Name	AI-Brand-Voice-Generator
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust (Python-based load testing) / Apache JMeter
Type of Testing	Load Testing & Stress Testing
Target Module / API	<code>/api/generate-copy</code> & <code>/api/extract-brand-voice</code>
Test Environment	Local Development Server (FastAPI + Uvicorn on localhost)
Test Date	12 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline Load Test: Single user generating content to establish benchmark latency.	1	60	Response time < 1.5s, 0% error rate.
2	Normal Load Test: Simultaneous content generation requests under typical usage.	10	120	Avg response time < 2s, error rate < 1%.
3	Peak Load Test: Simultaneous brand profile extraction and copy generation requests.	25	180	Avg response time < 3s, error rate < 1%.
4	Stress Test: Pushing concurrent requests beyond capacity to test system limits.	50	300	System handles traffic without crashing; error rate < 5%.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.42 seconds	Pass	Maintained low latency under normal user load.
2	Response Time (Max)	< 5 seconds	3.85 seconds	Pass	Spikes occurred during large text extraction requests.
3	Throughput (Req/sec)	> 10 req/sec	18.4 req/sec	Pass	Successfully processed concurrent requests.
4	Error Rate	< 1%	0.2%	Pass	Minimal rate-limit retries from external Gemini API.
5	CPU Utilization	< 80%	42%	Pass	Local FastAPI backend efficiently utilized cores.
6	Memory Utilization	< 80%	58%	Pass	Stable memory footprint during test duration.

Step 4: Observations & Analysis

- Latency Stability: The overall system maintained an average response time of 1.42 seconds, which successfully meets the target threshold of under 2 seconds.
- Third-Party API Dependency: The primary driver for peak response times (3.85s) was the network round-trip delay to the external Google Gemini API when processing large text prompts.
- Resource Efficiency: CPU and memory usage remained safely below the maximum 80% safety ceiling, reaching peak levels of 42% and 58% respectively under a 25-user concurrent load.
- Error Resilience: An error rate of only 0.2% demonstrates that FastAPI's asynchronous endpoint handling effectively manages incoming request queues without dropping connections.